

Lab-02 (Explanation)

Task-01 (a)

For this task I used nested loop to get the required sum and by using the nested loop it also fills up the condition of $O(n^2)$.

Task-01 (b)

For this task I used a single loop and some conditions to get the required sum and by using the single loop it also fills up the condition of $O(n)$.

Task-02 (a)

For this task I used a function called merged to merge the array and fulfill the requirements of $O(n \log n)$.

Task-02 (b)

For this task I used a while loop to merge two arrays into a singular array and also used the function append and extend and the time complexity of this function should be $O(n)$.

Task-03

In this task the logic is to determine overlapping, that can't be performed. So, in this task it checks each start time and the last process test's end time. If the current time is greater or equal to the last time then it adds.

Task-04

In this task the code attempts to iterate through each task and check the ability of a person to perform the task.

In this task there is no time constraint for this task 4. However, this problem can be solved by a sorting based approach $O(n \log n)$.